

Sample Project Documentation

Date	04 July 2026
Team ID	[Enter Team ID]
Project Name	HDI Predictor

Technical Overview

The **HDI Predictor** leverages machine learning algorithms to map 7 key socio-economic indicators directly to a country's Human Development Index (HDI) score.

Model Development & Performance

During the model selection phase, three algorithms were compared using 5-fold cross-validation RMSE:

Model Name	Cross-Validation RMSE	Test R ² score
Linear Regression	0.0152	0.984
Random Forest Regressor	0.0078	0.996
Gradient Boosting Regressor	0.0051	0.999

The Gradient Boosting Regressor was auto-selected and serialized as **best_model.pkl**.

Input Features & Range

Variable Name	Description	Value Bounds
Life_Expectancy	Life expectancy at birth (years)	30.0 to 90.0
Mean_Years_Schooling	Average years of education received by adults aged 25+	0.0 to 20.0
Expected_Years_Schooling	Number of years of schooling a child can expect	0.0 to 25.0
GNI_per_capita	Gross National Income per capita (PPP, inflation-adjusted)	100 to 150,000
Gender_Dev_Index	Ratio of female to male HDI	0.2 to 1.5
Gender_Ineq_Index	Composite measure showing loss in achievements due to gender inequality	0.0 to 1.0
CO2_per_capita	Carbon dioxide emissions per capita (production tonnes)	0.0 to 100.0

Flask Web Application Structure

The application uses Flask to serve a single-page style form interface. Submitting the form posts values to the **/predict** route, which calls **scaler.pkl** to scale values and evaluates the output using **best_model.pkl** before displaying the output tier classification.